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Research paper

Safety monitoring digital twin-based centralized model consolidation mechanism using dynamic node selection for multi-worker safety prediction

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ABSTRACT

The construction industry remains one of the most hazardous sectors, requiring innovative solutions to safeguard workers, especially in complex, dynamic outdoor environments. Digital Twin (DT) technology offers promising capabilities for real-time safety monitoring through virtual replicas of physical systems. However, existing DT frameworks rarely integrate comprehensive safety monitoring via a centralized model consolidation mechanism, such as Federated Learning (FL), which is explicitly tailored for multi-worker scenarios. Addressing this challenge, this paper proposes a FL-based Safety Monitoring Digital Twin (SMDT) framework designed to enhance multi-worker safety in resource-constrained settings. This enables real-time safety monitoring and control by representing on-site workers as virtual objects within a synchronized DT environment. A dynamic node selection mechanism based on client performance is employed to optimize global model convergence in FL. To validate the proposed approach, an edge computing-based experimental testbed using actual Raspberry Pi devices was implemented, using real-world construction safety data including worker status, weather conditions, and building structural parameters. Experimental results demonstrate the effectiveness of the proposed framework in significantly improving safety predictions and real-time monitoring efficiency. This research establishes a foundational work towards safer construction sites through intelligent, synchronized safety monitoring systems.

1. Introduction

The construction industry is one of the most hazardous sectors, characterized by dynamic and transient work environments where tasks and hazards constantly evolve. Non-fatal injury rates in construction surpass those in other industries by 71%, underscoring the critical need for innovative and effective safety solutions (Li et al., 2019). This sector presents a unique opportunity for technology-driven interventions due to its reliance on diverse processes, personnel, and equipment, coupled with the increasing availability of sensor-generated data. By leveraging data-driven management of physical systems, significant advancements in operational efficiency and safety can be achieved. DT technology, first introduced by Michael Grieves at a 2002 conference in Troy, Michigan Grieves (2019), has emerged as a transformative concept for creating digital replicas of physical systems, referred to as physical twins. These digital replicas enable the synchronization of data between physical and digital entities, allowing for optimization, simulation, and real-time monitoring of physical systems (Batty, 2021). With advancements in IoT, DT technology has been increasingly integrated into various phases of construction projects, including design, operation, and maintenance (Lo et al., 2021). During the construction phase, DT has demonstrated potential for real-time control and management of processes, equipment, materials, workers, and safety (Boje et al., 2020; Pan and Zhang, 2021).

In the context of safety, DT applications have shown promising results. For instance, DT-based systems have been developed to detect anomalies in the operational condition of assets, improving operational precision and reducing hazards (Lu et al., 2020b). Similarly, sensor-based safety management systems have been implemented to monitor

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Nomenclature

ML	Machine Learning
DT	Digital Twin
CL	Centralized Learning
FL	Federated Learning
DNN	Deep Neural Network
DNNR	Deep Neural Network Regression
IID	Independent and Identically Distributed
Non-IID	Non-Independent and Identically Distributed
IoT	Internet of Things
AI	Artificial Intelligence
AIoT	Artificial Internet of Things
DL	Deep Learning
MSE	Mean Squared Error
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination
LAN	Local Area Network
HTTP	Hypertext Transfer Protocol
TCP	Transmission Control Protocol

and track construction machinery operations within Cyber–Physical Systems (CPS), enhancing safety and situational awareness (Hasan et al., 2022). Despite these advancements, DT technology has not yet been fully realized in the construction industry for addressing worker safety challenges. Hou et al. (2020) highlighted that the lack of real-world implementation and the absence of a unified framework integrating diverse sensor data, processing technologies, and decision-making systems have limited the potential of DT. This fragmentation leads to inefficiencies in safety monitoring and operational decision-making (Teizer et al., 2022).

To bridge these gaps, this study introduces a novel framework that integrates FL with DT to enhance safety monitoring in multi-worker outdoor construction environments. FL offers a decentralized machine learning approach, enabling devices to collaboratively train models while preserving the privacy of locally stored data (McMahan et al., 2017). Unlike centralized learning, where data is aggregated at a central server, FL transmits only model updates from individual clients, significantly reducing communication overhead and addressing data privacy concerns (Li et al., 2021). This is particularly advantageous for resource-constrained IoT devices that generate continuous streams of time-series data, where constant data transmission to a central entity would be impractical (Wang et al., 2022).

In this study, a decentralized learning paradigm, FL, is employed to consolidate knowledge and form a global model through the collaborative training of multiple clients, which represent the multi-worker scenario, especially in outdoor occupational settings. Without needing to share the raw data, FL enables the construction of an optimal model that is used to monitor the safety of the workers. Moreover, a performance-oriented device selection method is employed to ensure the optimal performance of the learning model aggregation in every round. To do this, DT serves as a virtualization platform, where multiple devices can be dynamically registered and created as virtual entities to mirror the characteristics of their physical counterparts (Gao et al., 2023). This virtualization ensures synchronization between physical and virtual objects, enabling efficient participation in the FL process and facilitating real-time monitoring for worker safety. Fig. 1 illustrates the DT architecture, where the first layer comprises virtual twins of physical devices to establish a synchronized connection. The DT environment facilitates visualization of FL training processes, safety metrics, and performance predictions through an interactive web interface. Furthermore, the integration of DT enables monitoring and control to be conducted through HTTP, while model updates are transmitted via TCP (Pinyoanuntapong et al., 2022). To operationalize the framework, an edge computing system based on prior studies (Xu et al., 2022) is employed to communicate with sensor-based IoT devices and streamline data collection. A DNN (Liu et al., 2017) is collaboratively trained across FL clients and then deployed for inference.

The paper makes the following key contributions:

1. Proposes a novel federated learning-based safety monitoring digital twin (SMDT) framework that addresses current gaps in integrated safety monitoring systems and improves the safety of multiple workers in dynamic, outdoor construction environments.
2. Develops an integrated system that enables real-time safety monitoring and control while maintaining synchronization between physical and virtual devices, representing resource-constrained AIoT devices as virtual objects within a DT environment.
3. Introduces a dynamic node selection mechanism based on normal distribution to improve the convergence efficiency of global federated learning models, ensuring optimal client participation in each collaborative learning round.
4. Implements and experimentally validates the proposed framework on a physical testbed, using real Raspberry Pi edge devices and IoT sensors to collect and manage live construction safety data, demonstrating practical feasibility and performance in realistic settings.

The remainder of the paper is organized as follows: Section 2 presents a summary of related studies. The suggested FL-based SMDT for the construction worker safety inference system is presented in Section 3. Experimental and performance outcomes are described in detail in Section 4. Section 5 concludes the paper with a discussion of potential directions for future research.

2. Related work and background

The following subsections review the broad applications of DT technology, its utilization in construction safety, the integration of FL with DTs, and the use of advanced machine learning-based monitoring in industrial systems, which together form the conceptual foundation of our proposed SMDT framework.

2.1. General applications of digital twin technology

Digital Twin technologies have gained increasing attention across various industrial domains due to their ability to provide real-time synchronization between physical and digital entities. In structural engineering, DTs have been employed for assessing the seismic collapse resistance of infrastructure such as long-span cable-stayed bridges, helping to identify vulnerable components (Lin et al., 2021). In manufacturing, DT-driven integration frameworks have been developed to optimize performance evaluation in smart manufacturing enterprises, including the evaluation of green performance metrics (Li et al., 2022). Beyond engineering, DT concepts have expanded into the healthcare domain (Hassani et al., 2022), where human-centered digital twins are used to enhance personalized healthcare and wellness services. Recent research also explores Human Digital Twin (HDT) frameworks in Industry 5.0 environments, aiming to empower individuals and foster adaptive learning within cyber–physical–social ecosystems (Wang et al., 2024). Across all these sectors, the ability of DTs to integrate real-time sensing, virtual modeling, and predictive analytics has opened new frontiers for intelligent monitoring and control. Within the construction industry, DT adoption is accelerating (Jiang et al., 2021), yet comprehensive frameworks for real-time safety management remain relatively underexplored.

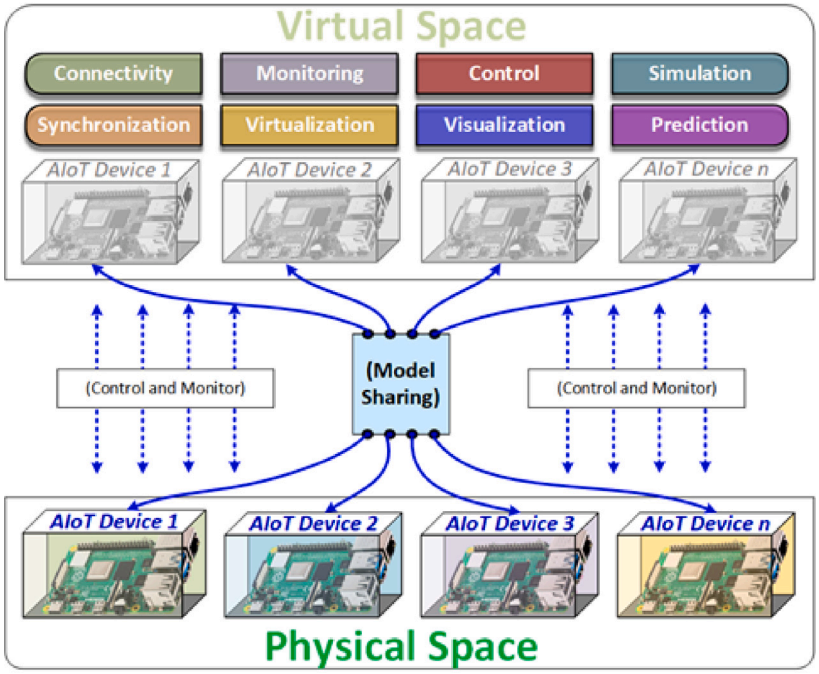


Fig. 1. Control characteristics of the virtual objects in digital twin.

2.2. Digital twin applications in construction worker health and safety

The construction sector, with its dynamic, hazardous work environments, presents a prime opportunity for applying DT-based solutions to improve worker health and safety. Several innovative systems have emerged in this space. For instance, IoT-integrated DT frameworks have been used for monitoring worker status and location in real-time using RFID and Bluetooth technologies (Zhao et al., 2021). Wearable sensor-based DT systems have also been designed to capture worker posture and motion through Inertial Measurement Units (IMUs), with real-time feedback enabled through AR (Akanmu et al., 2021). DT-based behavioral simulations have been developed to model worker responses to various safety interventions (Fang et al., 2023), providing valuable insights into optimizing safety training and protocols. Closed-loop DT systems for proactive safety control have also been demonstrated, using machine learning and case-based reasoning to guide safety decisions in ongoing projects (Sacks et al., 2020). Additionally, DT-based risk prediction models have been applied to specific construction tasks such as hoisting operations, where they help visualize and predict spatiotemporal safety risks in real-time (Liu et al., 2022). While these efforts demonstrate the potential of DTs for enhancing safety management, many current approaches are limited to single-use cases or small-scale implementations, lacking broader integration across multi-worker, outdoor environments.

2.3. Federated learning and digital twin integration

Recent years have seen growing interest in combining Federated Learning (FL) with DT to support privacy-preserving, distributed intelligence across diverse systems. Pioneering frameworks have been proposed to enable collaborative FL among city-level DTs for public health applications such as pandemic response (Pang et al., 2021). In industrial IoT environments, adaptive DT architectures have been designed to enhance FL by enabling device-specific modeling while reducing communication costs (Sun et al., 2020). Drone-based DT-FL frameworks have been introduced for privacy-aware data aggregation and storage in blockchain networks (Islam and Shin, 2023). Healthcare applications have benefited from hierarchical FL-DT models for anomaly detection with improved resilience to data poisoning and communication delays (Gupta et al., 2021). Other studies integrate FL with DT and mobile edge computing (MEC) to overcome latency and scalability challenges in heterogeneous network environments (He et al., 2022; Alhartomi et al., 2023; Lu et al., 2020a; AbdulRahman et al., 2023). Despite these advancements, existing works often focus on static or semi-static environments (smart cities, healthcare, IIoT), with limited attention to highly dynamic, mobile, multi-agent environments such as outdoor construction sites. Furthermore, many studies remain at the simulation stage, without extensive validation using real-world edge hardware.

2.4. Advanced machine learning-based monitoring in industrial systems

Advanced machine learning-based monitoring has become a central focus in the development of intelligent industrial systems, where real-time fault detection, anomaly recognition, and process optimization are critical. For instance, wavelet-entropy aided ML approaches have improved multiscale monitoring accuracy in chemical process industries (Ali et al., 2023). Integrated ML frameworks for physico-chemical process monitoring have shown strong performance in handling dynamic system behavior (Ali et al., 2024a). Distributed Canonical Correlation Analysis (CCA) with low-frequency control charts has enhanced statistical fault detection capabilities in complex industrial processes (Ali et al., 2024b). Likewise, dynamic ICA-distributed CCA methods (Ali et al., 2024c) and explainable fusion-based monitoring techniques (Ali et al., 2025) have contributed to improved interpretability and robustness in safety-critical applications. These trends highlight the increasing demand for intelligent, scalable, and interpretable monitoring frameworks across industrial domains. Our proposed SMDT framework draws on these concepts, applying federated learning, dynamic node selection, and DT-based explainable monitoring to the multi-worker construction domain — an area where such integrated approaches remain underdeveloped.

3. Proposed methodology

3.1. Dataset and model

A list of the sensor devices used for collecting environmental, structural, and worker-related data is shown in Table 1. The dataset includes

- Input dimension = 8
- Output dimension = 1
- Hidden layer 1 dimension = 64
- Hidden layer 2 dimension = 32
- Total parameters = 2689

Input 1	Load
Input 2	Strain
Input 3	Inclinometer
Input 4	Wind
Input 5	Temperature
Input 6	Precipitation
Input 7	Heart
Input 8	Location
Output	SafetyIndex

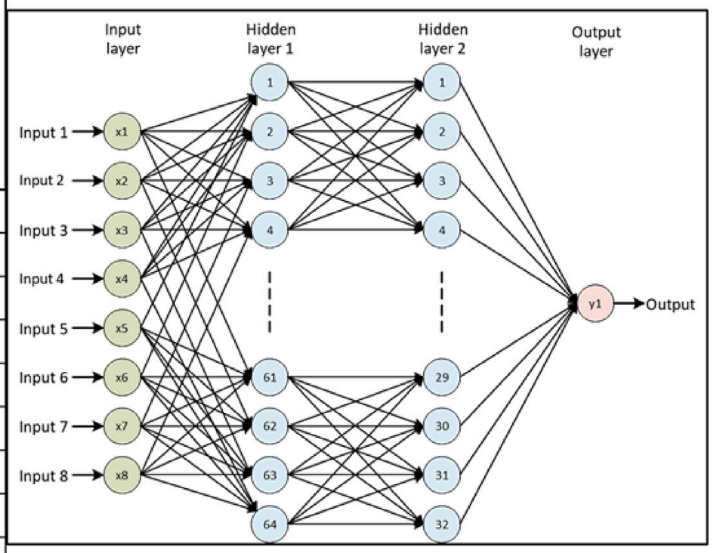


Fig. 2. DNN model architecture used for learning and prediction.

Table 1
Detailed experimental sensors descriptions for collecting construction environment data.

Sensing device	Feature	Description with units
Anemometer	Wind	Wind velocity (m/s)
Pluviometer	Precipitation	Rainfall height (mm)
DHT22	Temperature	Temperature at outdoor (°C)
Load Cell	Load	Force of pulling (kN)
Strain gauge	Strain	Variation in Length (μm/m)
Inclinometer	Inclinometer	Object's slope (°)
Heart rate sensor	Heart Rate	Heartbeat of human (BPM)
Handheld GPS	Location	Distance to the danger zone (m)

50,000 instances with eight features, normalized per node to accommodate sensor variability.

1. Weather condition: The weather on the construction site is an important consideration. Wind speed, precipitation, and extreme temperatures can all affect the safety of workers as well as the tools and materials used in construction (Karthick et al., 2023).
2. Building conditions: It is crucial to measure the load, strain, and inclination accurately into account to affirm the integrity of the structural and building's stability in an outdoor construction environment (Kang et al., 2023).
3. Worker status: To ensure the well-being of construction workers, it is essential to protect their safety and health. By tracking the heart rates of the workers, it is possible to learn about how their bodies are reacting to their workplace. Also, knowing where they are, especially in relation to the danger zone, is vital to guaranteeing their safety (Patel et al., 2022).

The provided Fig. 2 illustrates a DNNR model designed for training and predicting worker safety. This multilayered neural network is made up of an input layer, two hidden layers having 64 and 32 neurons each, and an output layer. It was built using a supervised learning methodology. The model utilizes various input parameters, including wind, temperature, precipitation, load, strain, inclinometer, heart rate, and location. The output of the model is the safety index, and it involves a total of 2689 parameters.

3.2. FL-based safety monitoring digital twin system

We developed an integrated framework combining DT and FL mechanisms to enhance the safety of construction workers. Fig. 3 illustrates the architecture of the proposed DT-based worker safety prediction system. The process begins with leveraging the core capabilities of the DT framework, including virtualization, synchronization, and visualization. Each AIoT device, representing an individual construction worker, is virtualized on the DT platform as a digital twin. This virtualization mirrors the physical device's characteristics to enable real-time monitoring and safety assurance. For this integration, physical devices must register with the DT platform by providing their port number and IP address to establish a connection.

In our proposed system, each worker's digital twin is constructed by combining multiple attributes captured in real time from edge devices. These include: (1) Worker Status — heart rate (biometric signal) and location relative to danger zones (via GPS); (2) Environmental Conditions — wind velocity, temperature, and precipitation; and (3) Structural Conditions — load, strain, and inclination of structural elements. Together, these attributes enable the dynamic representation of each worker's physical state and environment within the DT, supporting individualized safety monitoring and predictive analysis.

The FL mechanism is incorporated into the framework to collaboratively construct a globally optimal model while preserving the privacy of sensitive worker data. The FL server, embedded within the DT platform, facilitates both the initialization and aggregation of models. The initial model, constructed and tested on the DT platform, serves as the baseline, while the final model is obtained through iterative global aggregation of locally trained models across multiple devices. A DNN model is employed as the primary predictive model for assimilating worker safety information in outdoor construction environments. To expedite model convergence and improve performance, a dynamic device selection mechanism (explained in SubSection 3.4.1) is integrated during the FL server's aggregation process. This mechanism ensures efficient participation of devices in collaborative training.

The proposed framework utilizes communication protocols such as HTTP for real-time monitoring and control and TCP for secure model sharing. These protocols are implemented using lightweight web servers, with Flask managing monitoring commands and Django handling model update transfers. These technologies ensure efficient and scalable communication between the DT platform and the FL clients. Each client device is equipped with a DT assistance application, installed locally, which responds to control and monitoring commands from the DT platform. This application operates on the client's local IP address and registered port, maintaining synchronization with the DT platform. Additionally, the data across clients is assumed to be both

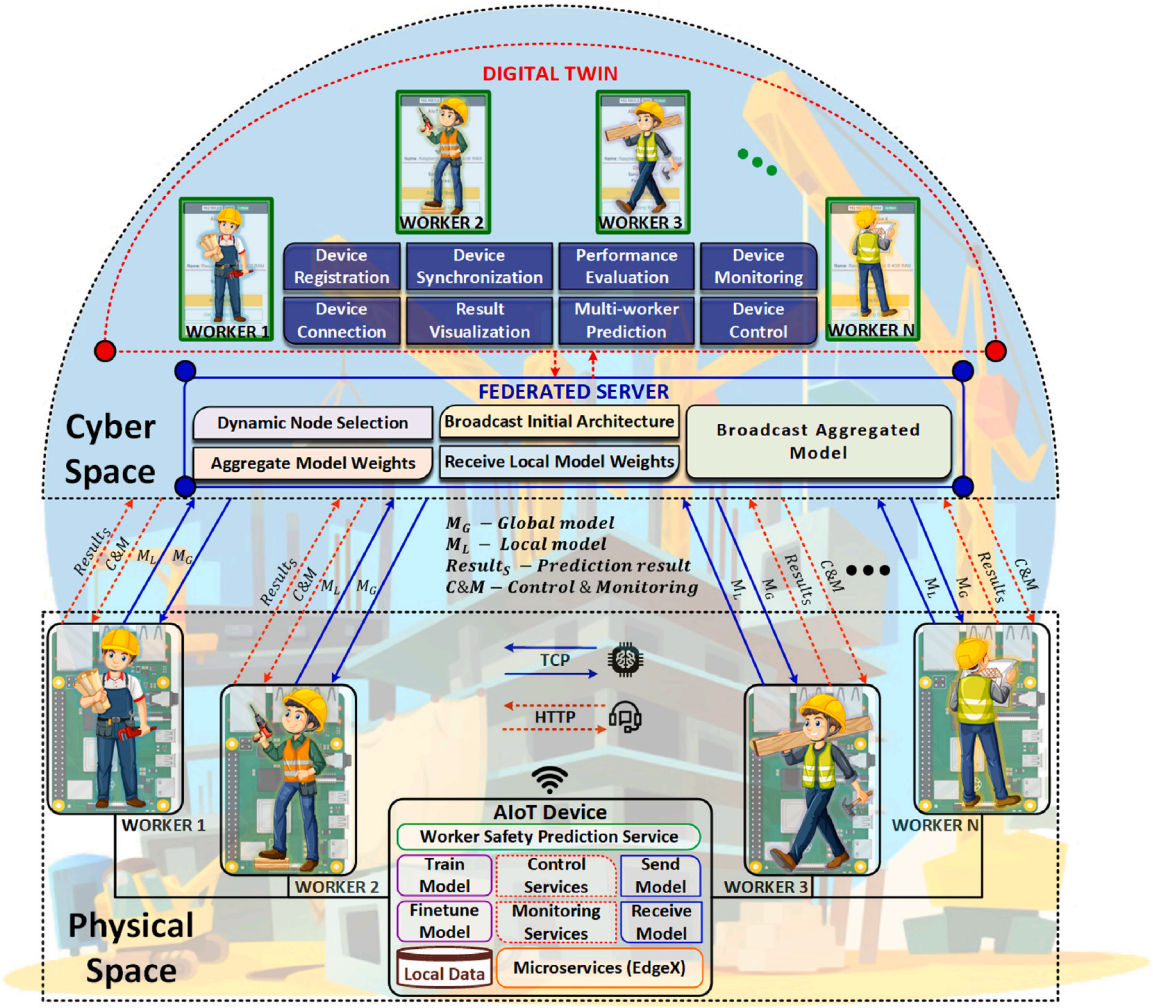


Fig. 3. Proposed federated learning-based safety monitoring digital twin for multi-worker safety prediction architecture.

IID and non-IID, ensuring that each client trains its model on data that is more closely aligned with the real-world scenario. The federated server shares the initial model architecture to enforce uniformity in the extracted feature matrix across all clients. This consistency in feature matrix dimensions ensures seamless aggregation of locally trained models into a single global model, thereby enhancing both the scalability and efficiency of the safety monitoring system.

The steps taken by the federated server and its clients are depicted in Algorithm 1, along with the node selection procedure and the model aggregation process that follows the receipt of updated models from each client.

Fig. 4 presents an integrated, four-tier architecture that underpins our SMDT, integrating with FL. At the base is the Construction Site Environment, where heterogeneous sensors capture real-time worker status (heart rate and GPS location), environmental conditions (wind, temperature, and precipitation), and structural parameters (load, strain, and inclination). These raw data feed upward into each AIoT Device, functioning as an edge node. Within each device, EdgeX microservices handle data acquisition and management, while control and monitoring services manage communication with the DT platform. The AI Module on each device adopts the global model architecture, fine-tunes it on the local data, and then extracts and sends updated model weights back to the federated server.

Above the AIoT layer, the Federated Server coordinates the FL cycle by preparing and broadcasting the initial model, receiving client updates, and aggregating them based on performance before redeploying a refined global model. At the top, the Digital Twin Layer provides three core functional spaces — Model Management (metadata collection, model creation, and layer setup), Monitoring Space (device registration, status monitoring, and metadata retrieval), and Control Space (parameter setting, device activation/deactivation, model broadcast, and inference control) — all supported by a central repository that stores device metadata and model performance. Communication uses HTTP for control and monitoring and TCP with JSON for model synchronization, enabling a seamless, secure, and transparent dashboard for visualization and system management.

3.3. Implementation of digital twin

The DT platform is implemented as a lightweight web-based interface accessible through any modern web browser (e.g., Chrome, Edge, Firefox), enabling wide accessibility for site managers, safety engineers, and IT staff without requiring dedicated client software. The system supports responsive real-time visualization of training progress, inference results, and worker safety indices with an update rate of approximately 1 s. Visualization efficiency is ensured through asynchronous REST APIs combined with WebSocket-based live updates, even when operating on resource-constrained devices such as Raspberry Pi clients. The platform also provides intuitive customization options, allowing users to dynamically register devices, configure FL training parameters (epochs, rounds), select models, and control which

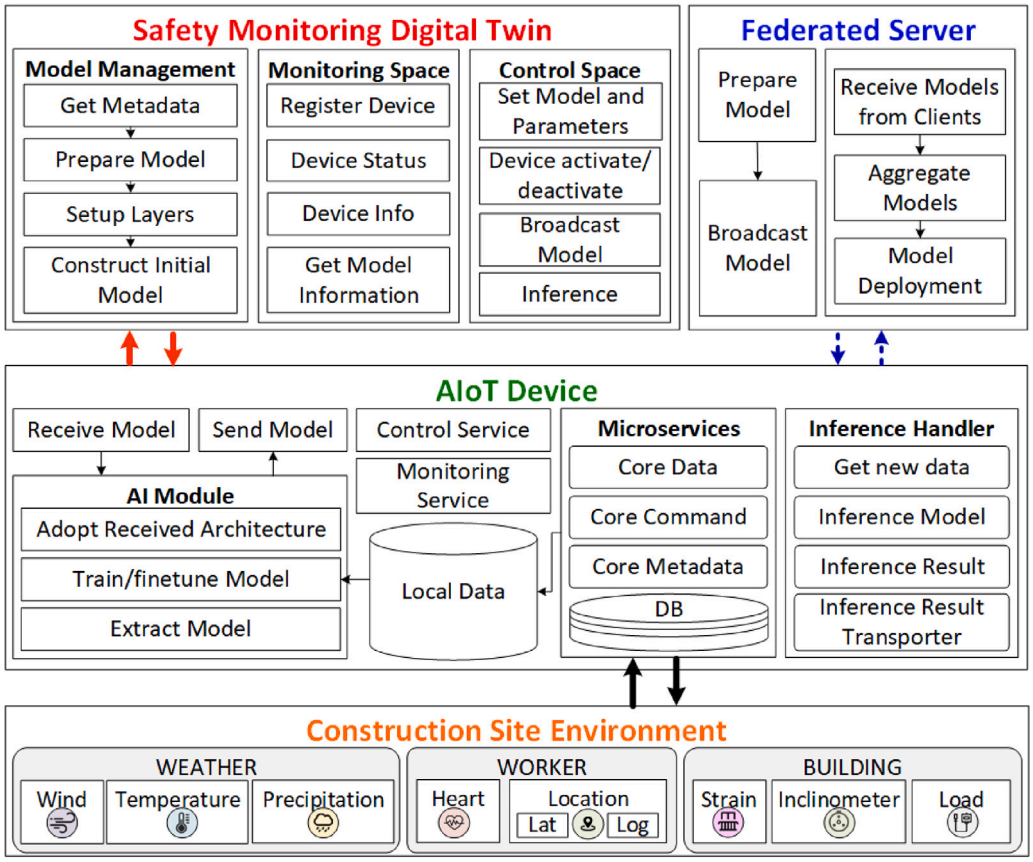


Fig. 4. Block diagram of the proposed SMDT system.

AIoT nodes and results are visualized. The modular UI structure supports future extensions for role-based access and stakeholder-specific dashboards.

3.3.1. Synchronization and management of data and model on DT

Fig. 5 depicts the conceptual diagram for data and model synchronization and management between the FL-based SMDT and its AIoT devices. These devices are created on the platform as virtual objects (VO) after registration. Model management system encompasses core functionalities: Model Visualization, which offers a graphical inspection of model architectures or metrics; Model Synchronization, ensuring alignment of various model versions across devices; Model Creation, the initial phase of defining new models; Model Update, which deals with refining existing models; and Model Deletion, eliminating obsolete models; Model Preparation, which indicates preliminary processes such as model initialization that takes user input to construct the model on DT.

The data in this study refer to the performance of the local update model in each round of collaborative training, as well as the features and data distribution of the sensitive local data. While local model performance is used in node selection for aggregation, local data features and their data distribution are considered when constructing the initial global model. Model Aggregation, central to FL, involves merging model parameters from diverse sources, as denoted by the formula $GM = \sum(LM1 + LM2 + LM3...LMn)$, which consolidates the weights of individual local models to create a global model. Model Weight Aggregation further emphasizes the specific focus on weights. Model Broadcasting distributes the finalized initial model or updated model parameters to individual nodes. At the base, local (LM1, LM2, LM3) and global model (GM) instances are depicted. The neural structures symbolize the architecture of individual local models, underlining their standalone nature in the federated framework.

The FedAVG is used for model aggregation at the server by assigning weights to the models based on their performance. Moreover, the performance of the received models is first analyzed; if the performance is below the threshold, the model will be excluded from the aggregation process for that specific iteration. In cases where the virtual and physical objects need to be synchronized, the synchronization property is crucial. Monitoring and controlling the FL training process is possible through DT, and the interactive web application of DT displays the prediction outcomes and performance metrics. TCP is mainly used for model synchronization, while HTTP is used for control and monitoring.

To facilitate the management, configuration, and monitoring of AIoT devices, the DT platform offers a web-based user interface (Fig. 6) that provides real-time visualization of safety indices, model training, and inference progress, while supporting device registration, model customization, and parameter tuning for various users and stakeholder needs. Each registered device is represented as an individual entity within the DT framework. The primary objective of this interface is to allow for the efficient monitoring and control of these devices in accordance with their designated functionalities. Users of the DT platform may initiate their experience by selecting the "Environment" option located at the top of the sidebar panel. Upon doing so, the system will present a list of registered devices along with their corresponding metadata, provided that the devices have already been registered. To configure the platform, DT users must first access the "Device Registration" section, which enables the registration of devices by entering their IP addresses and port numbers. Additionally, the "Model Management" feature plays a crucial role in configuring the learning model used in the FL process. The "Preparing Platform" function is designed to oversee registered devices that have successfully established a connection to the server following the initial registration phase.

On the DT platform, a node can dynamically register to participate in the collaborative training using FL. The following Fig. 7 depicts the

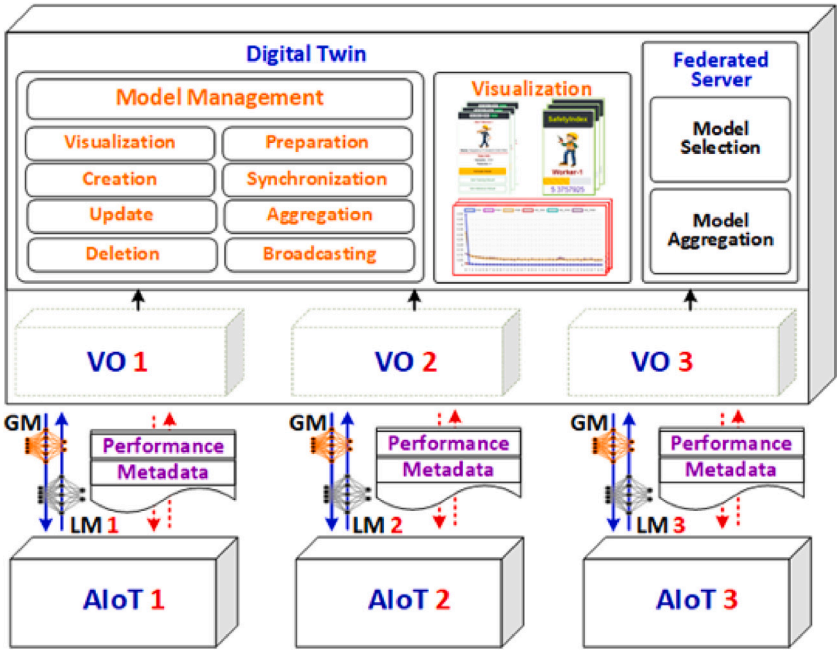


Fig. 5. Data and model synchronization and management on DT.

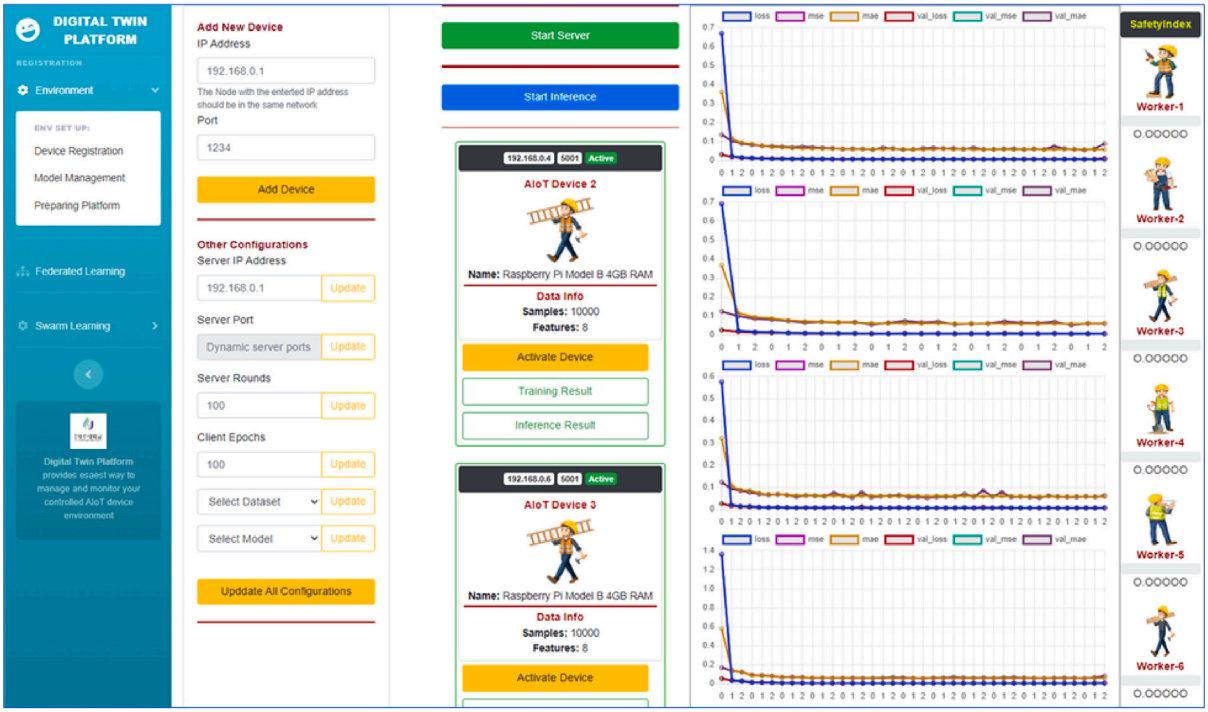


Fig. 6. Web-based UI of the DT platform, showing configurable options for device registration, training control, inference control, and real-time visualization of safety monitoring results.

registration interface on the DT platform's UI, followed by examples of registered devices' IPs and port information.

This (Fig. 8) interface on the DT platform provides configuration attributes for collaborative training in FL. The federated server house within the SMDT platform generates the configuration for every client. As seen in Fig. 8, the DT will create several JSON configuration files for the aggregator and nodes based on the user's configurations. The configuration comprises the name of the dataset, port, and IP of the server, local epochs, training rounds, name of the model, and the designated node's unique ID. To prevent collisions, the port of the aggregator — the federated server in FL — is dynamically assigned to each node. The aggregator identifies open ports and assigns one to each client based on their IP address and registered ports, before sending the configuration file to each client. Moreover, the FL parameter set by the DT user is also created as a JSON file to distribute to all registered clients.

The registration interface will appear when the user selects the device registration button in the DT sidebar. The device's IP address and port number are necessary for registration. The node with the specified IP address must be within the same network. In our testing environment, six devices are involved in collaborative training, and their respective IP addresses and port numbers are also presented for

Algorithm 1 Federated Learning with Nodes Selection

Input: Local dataset $E_1, E_2, \ldots, E_m$ on m client nodes, local model parameters $\{\theta_i\}_{i=1}^{M}$, the number of training rounds M, the number of local epochs N
Output: Updated parameters θ
Procedure *Client Node Executes:*
    for *each node* $i = 1, 2, \ldots m$ **in parallel do**
        Initialization
        current_round $\leftarrow 1$
        while current_round $\leq M$ **do**
            Update parameter θ_i with dataset E_i in N epochs
            Send training info to Digital Twin
            Get a list of selected nodes S_t from Digital Twin
            if $i \in S_t$ **do**
                Send θ_i to federated server
            end if
            Receive θ from federated server after aggregation
            $\theta_i \leftarrow \theta$
            current_round $\leftarrow$ current_round +1
        end while
    end for
end procedure
Procedure *Federated Server Executes:*
    current_round $\leftarrow 1$
    while current_round $\leq M$ **do**
        Receive a list of selected nodes S_t from Digital Twin
        Receive all weights $\{\theta_k\}$ from selected nodes
        $\theta \leftarrow$ aggregateWeights $(\{\theta_k\})$
        Send θ to all client nodes
        current_round $\leftarrow$ current_round +1
    end while
end procedure

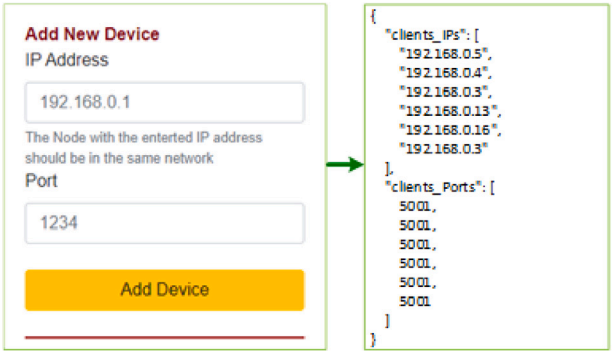


Fig. 7. Device registration interface on DT.

the convenience of DT users. The following Fig. 9 comes into sight when the DT user clicks on the "Model Management" button on the sidebar of the DT user interface. Preparing a model for FL clients is a key step that entails customizing the model's configuration through a user-friendly interface, as depicted in the figure.

Users can define various model attributes to fit their specific learning tasks and system requirements. These attributes include the model's name, loss function, activation function, architecture (number of hidden layers), optimizer, and input/output shapes. After configuring these parameters, users click the "Model Preparation" button to set up a fine-tuned model FL. This model is then distributed to client devices, allowing them to collaboratively train and enhance a global model while ensuring data privacy and decentralization. The interface for this process appears when a DT user clicks the "Model Preparation" button, as shown in Fig. 10. Here's a summary of the process:

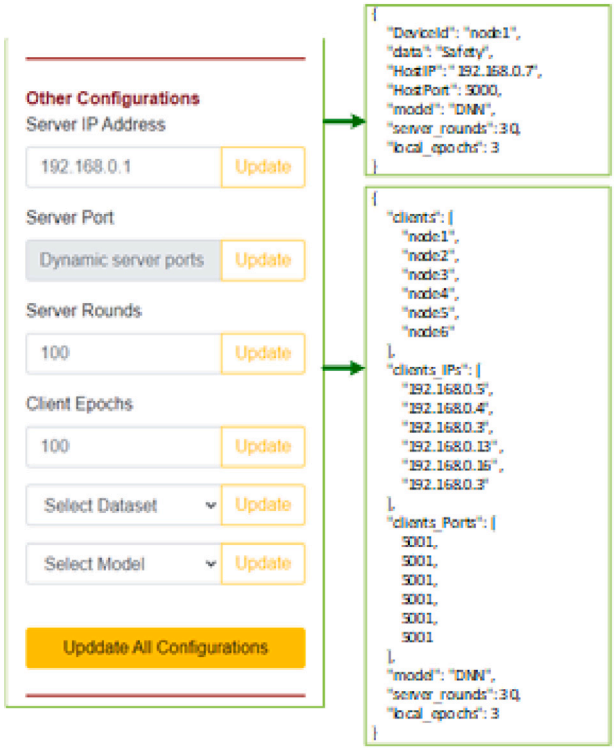


Fig. 8. Configuration interface on DT.

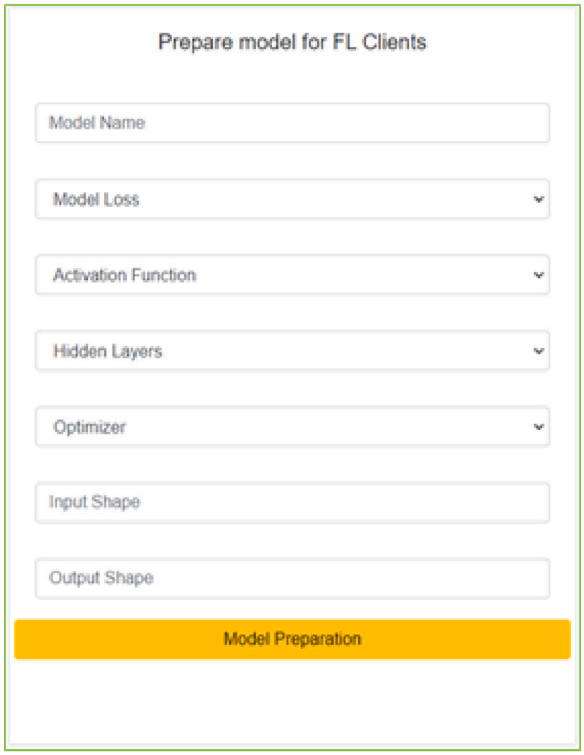


Fig. 9. Model architecture preparation.

- Model Definition: Users name their model (e.g., 'DNN') and specify its type.
- Loss Function: Mean squared error ('mse') is selected to evaluate model fit.

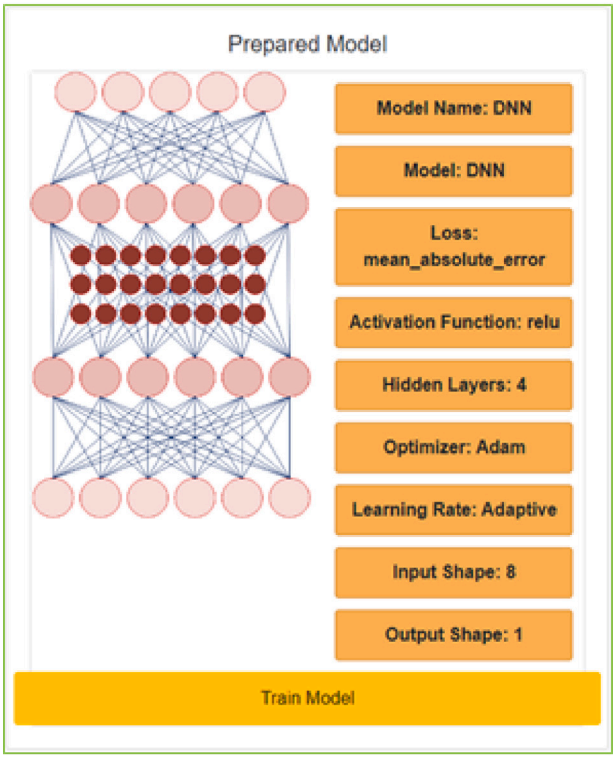


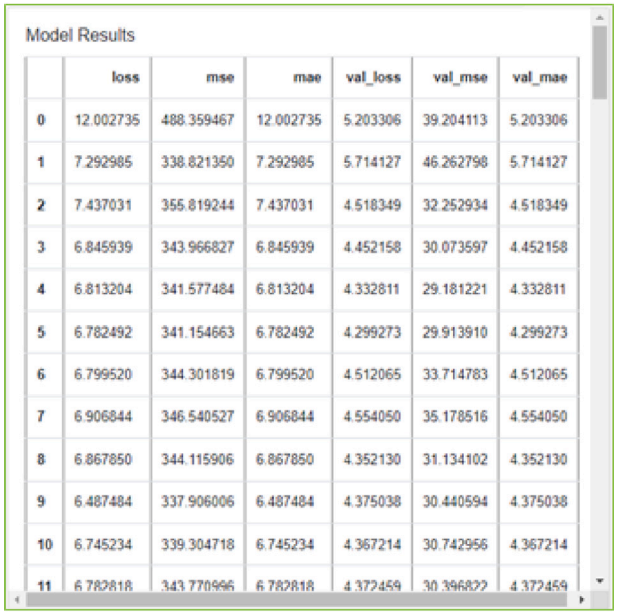
Fig. 10. Model preparation result on DT.

- Activation Function: 'ReLU' (Rectified Linear Unit) is chosen for non-linearity.
- Hidden Layers: Four hidden layers are defined for capturing complex data patterns.
- Optimizer Choice: 'Adam' is used for its adaptive learning rate.
- Input and Output Shape: The input shape is set to '8' and the output shape to '1'.
- Model Preparation: The user initiates 'Model Preparation' to create a custom model for FL clients, which they then train locally, preserving privacy.

When the DT user clicks the "Train Model" button, the interface shown in Fig. 11 appears, facilitating model training on the DT platform for distribution in the FL framework. The server computes performance metrics, which are displayed in a table, including 'loss,' 'mse' (Mean Squared Error), 'mae' (Mean Absolute Error), 'val_loss,' 'val_mse,' and 'val_mae.' This table distinguishes between training and validation metrics, enabling users to monitor the model's progress and performance and make informed adjustments.

The presented Fig. 12 becomes visible when the DT user selects the 'Preparation Platform' button on the DT user interface sidebar during configuration setup for the FL process. Upon registration of a device on the DT platform, related metadata like IP address, port number, hardware specifications, etc., is stored in the repository. Depending on the configuration set by the DT user, the devices are showcased to illustrate the communication topology within the network. In the displayed diagram, three devices are linked to the federated server but not directly to each other, exemplifying the collaborative training mechanism of FL.

Fig. 13 provides an overview of the control buttons for learning operations, inference, and virtual replicas of AIoT devices on the SMDT platform. Here are the key functionalities of these buttons:



	loss	mse	mae	val_loss	val_mse	val_mae
0	12.002735	488.359467	12.002735	5.203306	39.204113	5.203306
1	7.292985	338.821350	7.292985	5.714127	46.262798	5.714127
2	7.437031	355.819244	7.437031	4.518349	32.252934	4.518349
3	6.845939	343.966827	6.845939	4.452158	30.073597	4.452158
4	6.813204	341.577484	6.813204	4.332811	29.181221	4.332811
5	6.782492	341.154663	6.782492	4.299273	29.913910	4.299273
6	6.799520	344.301819	6.799520	4.512065	33.714783	4.512065
7	6.906844	346.540527	6.906844	4.554050	35.178516	4.554050
8	6.867850	344.115906	6.867850	4.352130	31.134102	4.352130
9	6.487484	337.906006	6.487484	4.375038	30.440594	4.375038
10	6.745234	339.304718	6.745234	4.367214	30.742956	4.367214
11	6.782818	343.770996	6.782818	4.372459	30.396822	4.372459

Fig. 11. Evaluation interface of the prepared model.

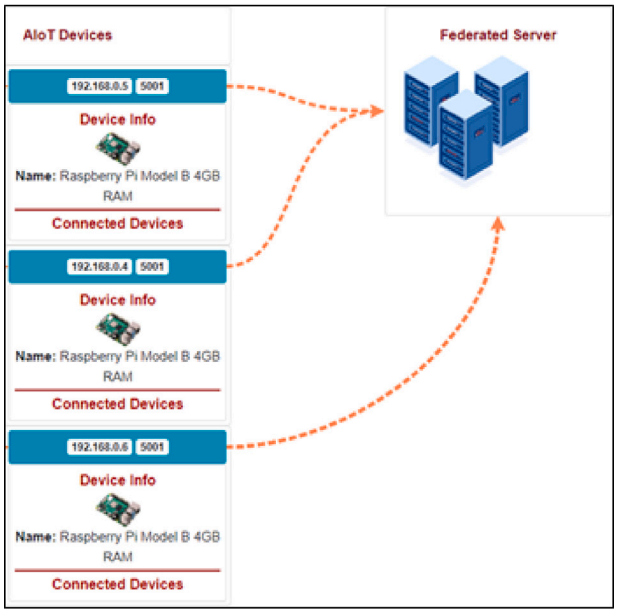


Fig. 12. Visualization of communication network for FL-based SMDT.

- Device Activation: This feature allows users to activate or deactivate individual nodes, which are AIoT devices. Activation establishes communication between the aggregator and its associated nodes.
- Training Control:
  - Start Server: DT users can initiate the training process by clicking this button. The process begins after the DT user configures the FL parameters.
  - Get Training Result: This button provides real-time access to training progress and results. Users can monitor various metrics, view graphs, and access statistics related to the training process.

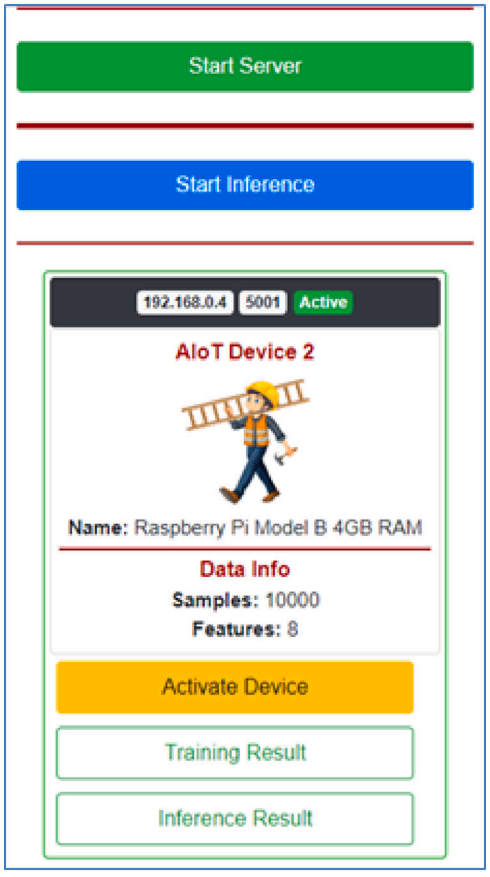


Fig. 13. Virtual replica of a registered device on DT.

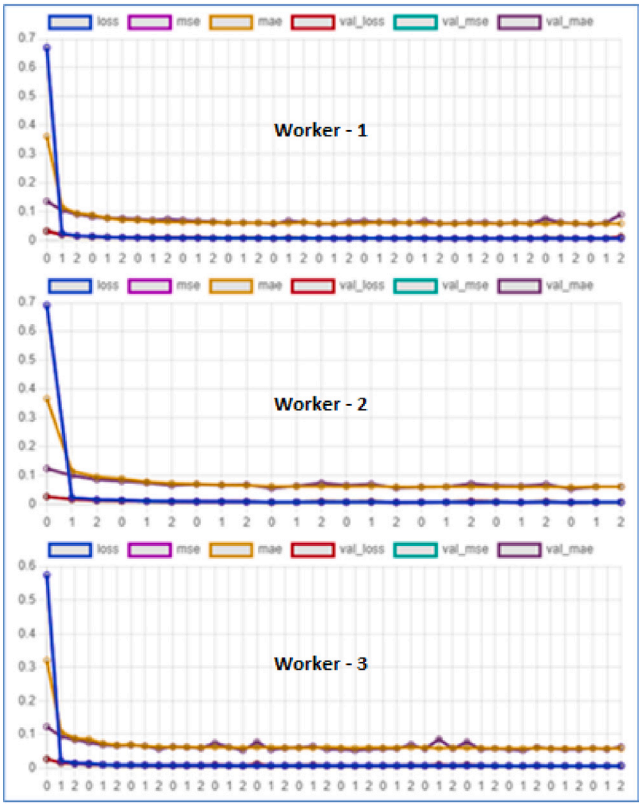


Fig. 14. Visualized model evaluation result of the AIoT devices.

- Inference Control:
  - Start Inference: Users can trigger inference processes on specific edge devices. Inference involves using trained ML models to make predictions or decisions.
  - Get Inference Result: The results of inference processes are displayed on the DT user interface, offering real-time access to prediction outcomes for worker safety.
- Data Visualization: The DT user interface displays data (metadata) collected from AIoT devices using charts, graphs, or maps. It also shows the results of training and inference processes. These visualizations provide a clear and informative way to understand and interpret the data and outcomes.

The results of the assessment of the model performance for workers 1, 2, and 3 — represented by AIoT devices 1, 2, and 3 — are shown in the following Fig. 14. For the convenience of DT users, the performance of the learning model on each node can be visually evaluated. This is achieved by having each participant in collaborative machine learning send their model performance results in CSV format to the DT platform, which is then compressed into JSON format. On the DT platform's user interface, the training model results from the AIoT devices are displayed. These metrics provide valuable insights into the performance of the learning models, aiding in assessment and decision-making.

3.4. Implementation of federated learning

3.4.1. Dynamic node selection algorithm for enhanced model consolidation

Improving the global model's overall performance depends critically on the convergence of the local models. We use FedAVG as the server-side aggregation method in our approach. First, clients carry out local updates after the FL server distributes the initial model containing randomly initialized weights. ($LU \geq 1$)

$$w^k_{t+i+1} = w^k_{t+i} - \alpha_{t+i} \frac{\partial error}{\partial w^k_{t+i}}, i = 0, 1, \ldots, LU = 6, \tag{1}$$

where k represents the number of clients and t denotes the cumulative count of local updates completed in earlier rounds of communication.

If k (FL's client) comprises local data instances: $x_{k,1}, x_{k,2}, \ldots, x_{k,n_k}$, subsequently, the regional loss of a specific client is described as:

$$P_k = \sum_{j=1}^{n_k} l(w^k_{t+i+1}; x_{k,j}) \tag{2}$$

where $l(.;.)$ represents a user-defined loss function, which is used to assess the client's performance k. Weights forwarded by clients w^k_{t+LU} to the server, and these weights are aggregated by the server to create an updated global model $w^1_{t+LU}, \ldots, w^n_{t+LU}$. When it comes to client participation, the server primarily conducts two types of aggregation:

1. **Full Node participation**. All registered clients' weights are combined in this mode of aggregation, and the performance of individual clients is ignored.

$$w_{t+LU} = \frac{1}{N} \sum_{k=1}^{N} w^k_{t+LU} \tag{3}$$

   Unfortunately, all device participation has a major impact on the global model. For example, in a twelve-client FL network, the global model could be adversely affected by any two clients participating if their performance is poor.
2. **Dynamic Node Selection**. Clients selected on the basis of various performance metrics are involved in the dynamic node selection scenario. Choosing the optimal models for aggregation

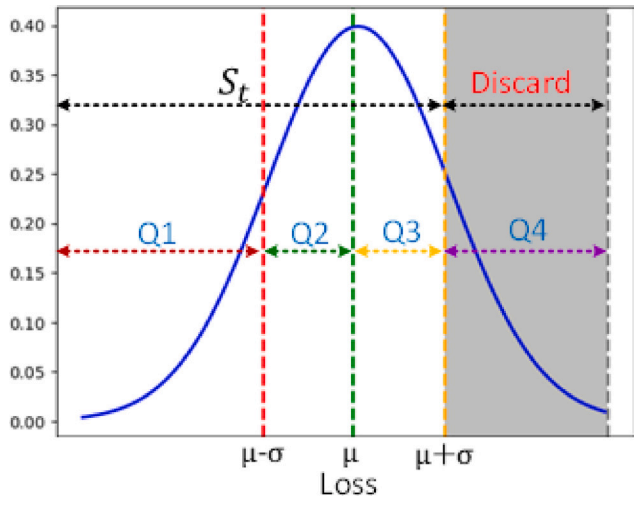


Fig. 15. Performance of the clients based on normal distribution.

makes this approach more realistic. By combining the weights of the chosen clients, the global model is produced.

$$w_{t+LU} = \frac{1}{count(S_t)} \sum_{k \in S_t} w^k_{t+LU} \tag{4}$$

where, for tth server round, S_t represents the set of the clients that are selected.

In this study, a node selection mechanism based on a normal distribution (Fig. 15) is employed.

$$f(p_k) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{p_k-\mu}{\sigma})^2} \tag{5}$$

where,

$\mu = \frac{1}{N}\sum_{k=1}^{N} p_k$ the mean of every client's performance.

$\sigma = \sqrt{\frac{\sum(p_k-\mu)}{N}}$ the standard deviation of each client's performance.

As shown in (Fig. 15) and stated in the following Eq. (6), the performance (loss) distribution is computed for each server round, and clients are selected if their loss falls below a predetermined level.

$$S_t = \begin{cases} \text{select} & \text{if } p_k \in [Q_1, Q_2, Q_3] \\ \text{discard} & \text{otherwise} \end{cases} \tag{6}$$

Aggregation is carried out after the node selection procedure, and the global model that is produced is subsequently shared with each and every registered client.

3.4.2. FL-based collaborative training

Fig. 16 shows the collaborative learning process based on the FL mechanism. In our experiment, six devices (Raspberry Pi 4B) are employed to evaluate this approach. Additionally, two types of experiments are conducted, such as full device participation and the dynamic node selection method. On each device, local data collected from the respective emulator is stored in CSV format. The following are the steps in each training round of the FL process.

1. Initialization: In the first round, the federated server distributes the global model weights as the starting point.
2. Local Training: For subsequent rounds, the federated server sends the current global model weights to the participant nodes. These nodes then perform local training using their own data, which has been preprocessed and transformed for compatibility with the learning model.
3. Parameter Aggregation: After local training, each node sends its updated model parameters to the server. Since each node uses different data, these parameters will vary.
4. Model Update: The federated server aggregates these parameters, using FedAVG as the primary approach, to create an improved global model.
5. Convergence: This process repeats from step 1 until the model converges.
6. Deployment: The trained global model is sent to every edge node that participated, once it has converged.

Table 2
Comprehensive overview of FL-based safety monitoring digital twin testing environment.

Entity	Hardware	Softwares
Digital twin/Federated server	PC, Windows 10 Pro	Python 3.9, HTML 5, CSS 3, JavaScript, Keras 2.6, TensorFlow 2.6, Flask 2.2.2, Django 4.2.7, MySQL Workbench 8.0
AIoT device	Raspberry Pi 4B, Ubuntu 20.04 Server	Python 3.8, Keras 2.6, TensorFlow 2.6, Flask 2.2.2
Emulator	Raspberry Pi 4B, Ubuntu 20.04 Server	Go 1.18.5, Docker 20.10.21, Docker-compose 1.25.0, EdgeX (Ireland)

The current system was implemented and evaluated on a physical testbed of Raspberry Pi 4B devices, simulating multi-worker conditions, using real-world construction safety data provided by a Korean research institute. Field deployment at active construction sites is planned for future stages of this research.

4. Experiment and results

The proposed SMDT framework was implemented and validated on a physical testbed consisting of Raspberry Pi 4B devices as AIoT clients, connected to a DT platform and FL server running on a high-performance PC. Real-world construction safety data provided by a Korean research institute was used for model training and validation. This section presents experimental results demonstrating model accuracy, convergence, inference latency, and system performance.

4.1. Experimental environment

To evaluate the proposed SMDT framework and the centralized model consolidation mechanism based on FL, we conducted extensive experiments using a practically implemented testbed (Fig. 17) consisting of both physical devices and edge computing components. Table 2 provides a detailed overview of the hardware and software environments used for the DT/FL server, AIoT devices, and IoT emulators.

In this testbed, the Federated Learning process is performed across multiple Raspberry Pi devices, which act as FL clients representing on-site workers and resource-constrained AIoT devices. These devices are deployed to collect real-world construction safety data, including worker status, environmental conditions, and structural parameters, to train a DNN model for effective safety inference in outdoor construction environments. The goal is to develop an optimal model capable of providing real-time safety monitoring and risk predictions for workers. Unlike purely simulated environments, this experimental setup integrates actual Raspberry Pi hardware with edge-based data collection, enabling us to validate both the practicality and reliability of the proposed system in a dynamic, realistic context. The DT platform and FL server run on a high-performance desktop computer to manage model aggregation, coordination, and visualization tasks.

Fig. 18 illustrates the overall architecture of the experimental environment used to test the SMDT framework for multi-worker safety monitoring in outdoor construction settings. The FL server and DT platform run on a desktop PC with Windows 10 Pro, powered by an Intel Core i7 processor (2.5 GHz), 64 GB of memory, and 1 TB of

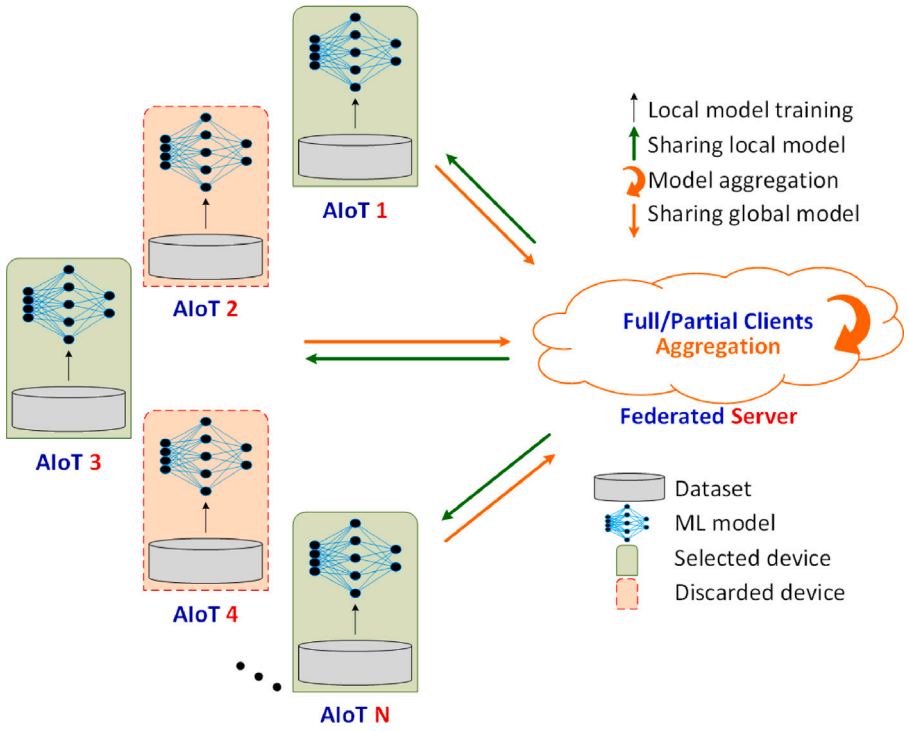


Fig. 16. Collaborative training process based on FL.

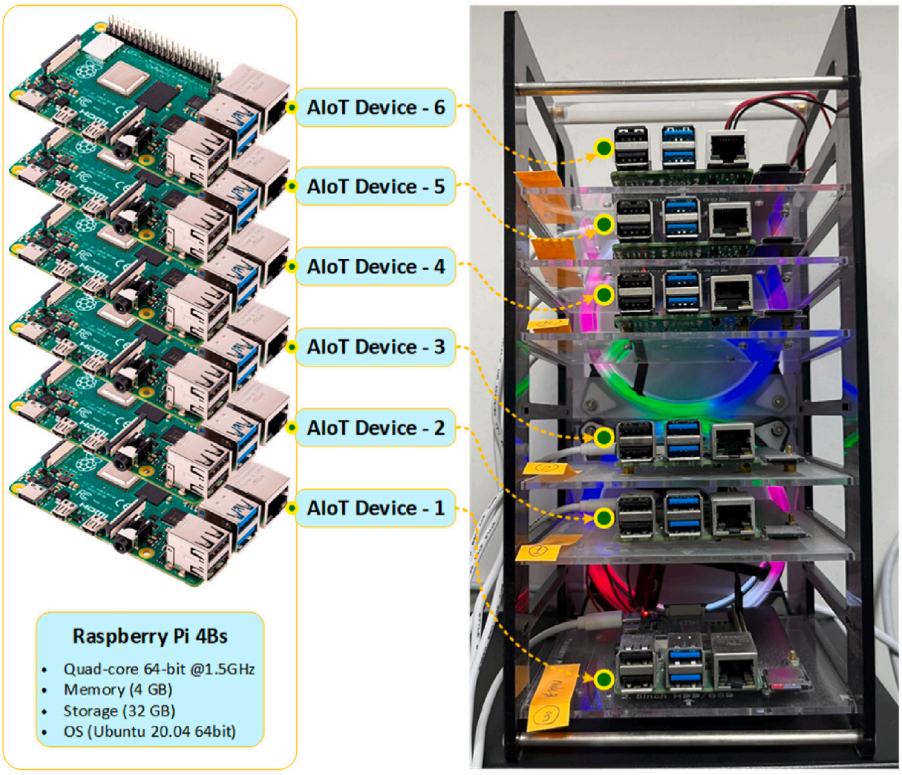


Fig. 17. Physical testbed of Raspberry Pi 4B devices used as FL client nodes.

storage. The AIoT devices and emulator nodes are based on Raspberry Pi 4B boards (quad-core 1.5 GHz 64-bit CPU, 4 GB memory, 32 GB storage) running Ubuntu 20.04 server OS. All devices in the testbed are connected through a dedicated LAN, with the server connected via Ethernet and Raspberry Pi devices connected via Wi-Fi through a router. This architecture simulates the communication and computation constraints of actual construction site deployments and supports practical validation of model training, synchronization, and safety inference.

Although the testbed uses real-world data from actual construction environments (provided by a Korean research institute), the current

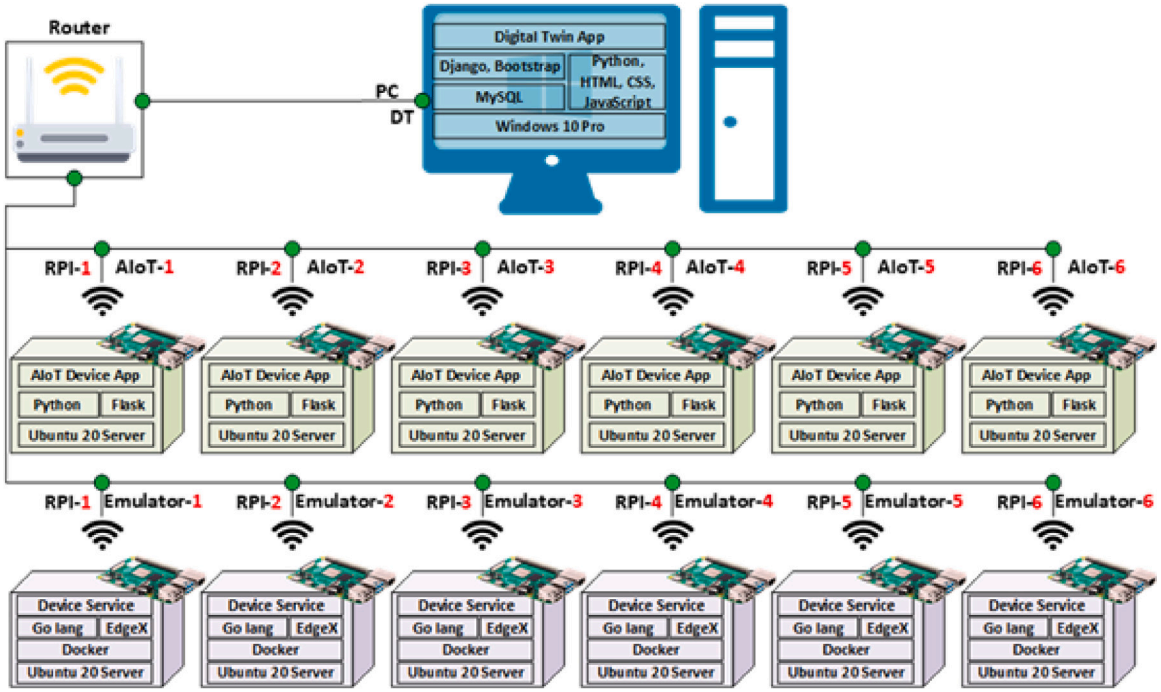


Fig. 18. Overall experimental environment architecture.

experiments were conducted in a controlled testbed environment using Raspberry Pi devices to emulate on-site conditions. Field deployment at an operational construction site is planned as future work.

4.2. Worker safety prediction

Fig. 19 illustrates the inference process deployed on the edge nodes to perform real-time worker safety prediction. In this system, sensor data from the outdoor construction environment is collected every 15 s by an emulator simulating field conditions. The data stream, including structural (strain, load, inclinometer), environmental (wind, temperature, precipitation), and worker-specific (heart rate, GPS location) attributes, is transmitted via EdgeX microservices and stored in the local repository. After preprocessing, using the same normalization applied during model training, the test data is passed through the deployed inference model on each edge device. The model outputs a Safety Index, representing the predicted safety level of the corresponding worker in near real-time. This process enables continuous monitoring and prediction of worker safety with low latency at the edge, without requiring centralized computation, making it suitable for dynamic outdoor construction sites.

4.2.1. Inference results and real-time performance

Furthermore, Fig. 20 displays the relationship between the target and predicted indexes of worker safety for six nodes using the best-predicted model obtained from the collaboratively trained model. It makes sense to gather data points for forecasting every fifteen seconds, with 100 data observations in dangerous settings like outdoor spaces where construction takes place. To increase worker safety, it is advisable to take into account shorter prediction time intervals. This figure shows a close match between the predicted and actual worker safety values 25 min ahead in an outdoor construction site.

Fig. 21 presents the average delay time observed by all nodes when predicting worker safety indices using real-time data streamed from the emulator. Since FL deploys the trained model on nodes after training completion. Consequently, the prediction process becomes significantly faster compared to conventional prediction techniques commonly employed in centralized machine learning architectures. After running the test 20 times, we were able to get an average time of 0.551 s. In the experiment, the shortest and longest delay times are 0.529 and 0.580 s, respectively. Every process in our system has a fast time delay that is fully functional in real-world scenarios.

4.3. Performance analysis

This section evaluates the system's performance in terms of training time, latency, and model accuracy under both IID and non-IID data distributions.

4.3.1. Training time and latency

Fig. 22(a) compares the total training time (in seconds) per communication round between full FL and dynamic node selection FL. The experiment was conducted using six Raspberry Pi 4B devices with limited resources. As shown in the figure, FL-dynamic requires slightly more time per communication round at later rounds due to the additional step of evaluating client model performance for node selection, while FL-full incurs a greater aggregation overhead from processing updates from all clients. Despite these differences, the overall training delay between the two approaches remains practically negligible, indicating that the node selection mechanism does not introduce significant latency or system bottlenecks.

Fig. 22(b) shows the average per-client training time across 30 communication rounds, along with the number of training samples per node. It is evident that clients with larger data volumes require longer local training times, and since the FL server must wait for all participating nodes to complete training before aggregation, slower nodes can contribute to overall delay. This further supports the rationale for dynamic node selection: by adapting to client performance, it can help mitigate potential bottlenecks caused by slower or unbalanced nodes.

We used a number of statistical evaluation metrics (Rainio et al., 2024), including MAE, RMSE (Hodson, 2022), and R^2 (Steyerberg et al., 2010), to evaluate the performance of our model in this work.

- MAE: A popular evaluation metric that measures the difference between a variable's actual and predicted values. It is computed

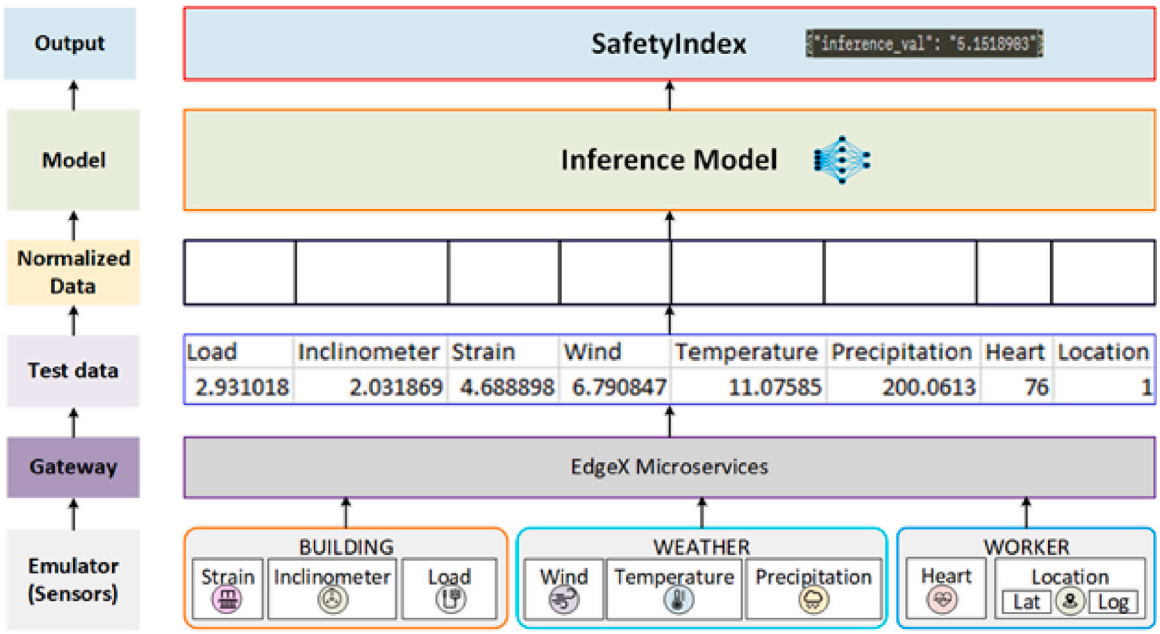


Fig. 19. Worker safety prediction process on the node.

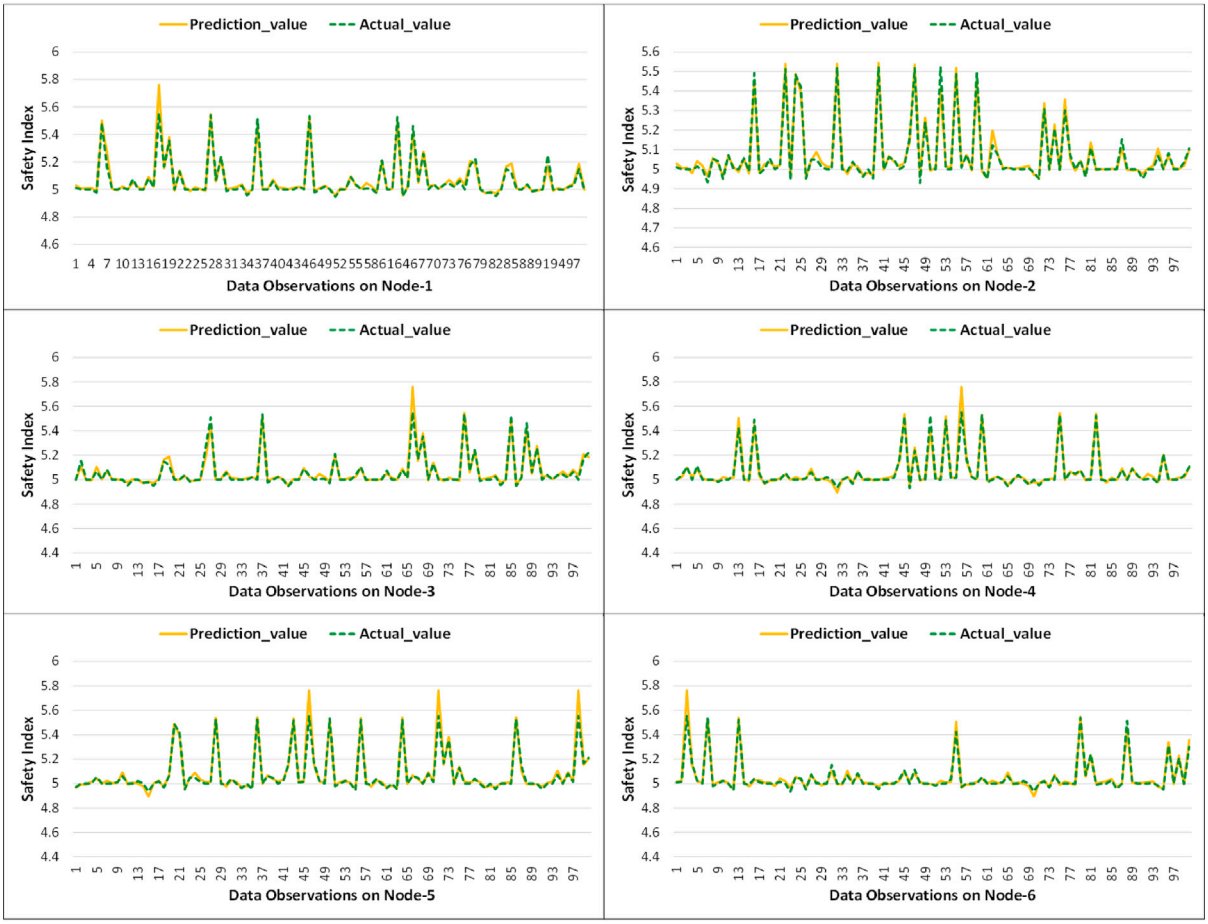


Fig. 20. Worker safety index: Actual versus predicted on all nodes.

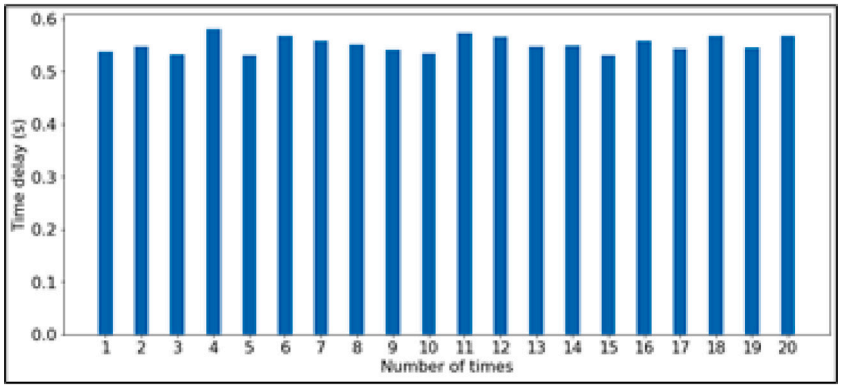
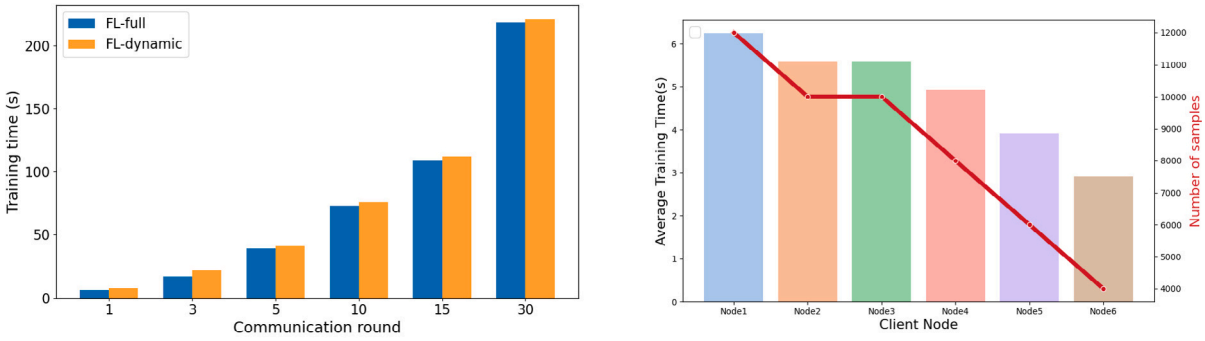


Fig. 21. Average delay time of all nodes for 20 inference times.



(a) Training time comparison across communication rounds. **(b)** Average client training time vs. data volume.

Fig. 22. Training time and latency analysis: (a) FL-full vs. FL-dynamic across rounds, (b) per-client training time based on data volume.

as the absolute distinction between the target and predicted values. The Equation for it is in (7):

$$MAE = \frac{\sum_{i=1}^{n} \left| Y_i - \hat{Y}_i \right|}{n} \tag{7}$$

- RMSE: This measure is employed to assess the error rate of the regression model and verify that the error values closely match the objectives. The equation that follows is used to calculate the Mean Square Error (MSE) square root (8):

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} \left(Y_i - \hat{Y}_i \right)^2}{n}} \tag{8}$$

- R^2: A statistical measure that signifies the proportion of variability in the dependent variable that can be accounted for by one or more independent variables. The equation for R2 is (9):

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} \tag{9}$$

We evaluate our six-client FL setup on a 50,000-sample worker safety dataset with 8 covariates and one continuous target variable under both IID and non-IID data partitioning strategies. For the IID split, we shuffle the entire dataset and assign an equal number of 10,000 samples to each client by ensuring that the distribution of the target variable is nearly identical across all clients. This is visually confirmed in Fig. 23, where the kernel density estimates (KDEs) of the target variable for all six clients almost perfectly overlap, which indicates a uniform distribution.

To create a more realistic non-IID scenario, we discretize the continuous target into ten quantile-based bins and, for each bin, sample a probability vector p from a Dirichlet distribution ($\alpha = 0.5$) to induce moderate heterogeneity. Within each bin, samples are allocated to

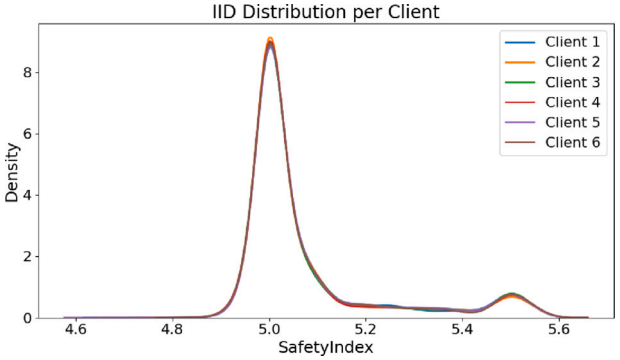


Fig. 23. IID distribution per client.

the six clients according to p, resulting in each client receiving a distinct proportion of the overall target distribution. As illustrated in Fig. 24, this Dirichlet-based partitioning results in highly heterogeneous and skewed local data distributions among clients, closely mimicking the non-IID data heterogeneity typically observed in actual federated learning deployments. Despite the diversity at the client level, the global data statistics are preserved.

4.3.2. IID setting

Fig. 25(a) and (b) illustrate the per-node validation loss over 30 communication rounds for both the FL-dynamic (dynamic node selection) and FL-full (full node participation) strategies using private worker safety data under an IID setting. Each curve represents the loss trajectory for a separate client node. Both approaches demonstrate rapid convergence within the initial rounds, with all nodes achieving

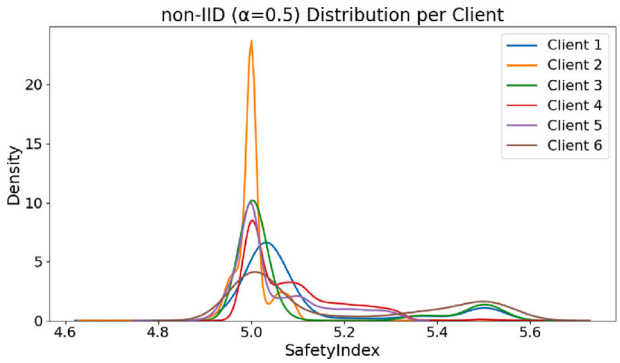


Fig. 24. nonIID distribution per client.

Table 3
Model performance and training time under IID setting.

Method	MAE	RMSE	R^2 Score	Training time (s)
CL	0.0451	0.0627	0.9748	198.2
FL-full	0.0548	0.0708	0.969	222.4
FL-dynamic	0.0478	0.0582	0.978	224.9

low and stable loss values after approximately five rounds. However, the FL-dynamic approach generally yields slightly lower loss values across nodes compared to FL-full, suggesting that performance-based node selection can enhance convergence speed and model generalization, even under IID data conditions.

Fig. 26 presents a comparison of three evaluation metrics, MAE, RMSE, and R^2 score, for FL with full participation and FL with dynamic node selection across multiple communication rounds. As shown in Fig. 26(a), the MAE decreases progressively as the number of rounds increases, with dynamic node selection consistently achieving lower error rates and faster convergence. A similar trend is observed in RMSE 26(b), where dynamic node selection outperforms full participation by enabling more stable and efficient learning.

Finally, the R^2 scores (Fig. 26(c)) further confirm this pattern: while both methods improve over time, dynamic node selection achieves consistently higher R^2 values, indicating better model generalization. These results demonstrate that performance-aware node selection can enhance learning efficiency and prevent global model degradation that may occur when poorly performing nodes are included indiscriminately in each aggregation round.

Table 3 presents the performance of the proposed safety prediction framework under IID data settings. The dynamic node selection approach outperforms FL-full in all evaluation metrics, achieving lower MAE and RMSE, as well as a higher R^2 score. Compared to centralized learning (CL), FL-dynamic offers a privacy-preserving alternative with competitive accuracy. The additional training latency introduced by the node selection mechanism is minimal (only +2.5 s compared to FL-full), confirming that the performance benefits are achieved without a significant time overhead.

4.3.3. Non-IID setting

Fig. 27(a) and (b) illustrate the per-node learning curves under non-IID data settings for FL with full participation and FL with dynamic node selection, respectively. Similar to the IID case, both approaches exhibit rapid convergence within the first few communication rounds. However, in the non-IID scenario, the impact of node selection is more pronounced: dynamic node selection (Fig. 27(b)) achieves slightly faster convergence and more consistent loss reduction across all nodes. In contrast, FL with full participation (Fig. 27(a)) shows slightly more fluctuation in later rounds. These results suggest that performance-aware node selection is particularly beneficial in heterogeneous data environments, where variations in local data distributions can otherwise slow down global model convergence.

Table 4
Model performance and training time under non-IID setting.

Method	MAE	RMSE	R^2 score	Training time (s)
FL-full	0.0603	0.934	0.9264	221.3
FL-dynamic	0.0578	0.0820	0.9650	224.9

Fig. 28 shows the comparative evaluation of FL with full participation versus dynamic node selection under non-IID data settings, using three key metrics: MAE, RMSE, and R^2 score. As seen in Fig. 28(a), the MAE values for all nodes are consistently lower when using dynamic node selection, indicating more accurate local model performance. Similarly, RMSE results in Fig. 28(b) demonstrate that dynamic node selection yields lower root-mean-square error across nodes, highlighting better generalization of the global model.

Finally, the R^2 scores plotted in Fig. 28(c) show that dynamic node selection leads to faster improvement and higher final R^2 values as communication rounds increase, confirming enhanced learning efficiency in non-IID scenarios. These findings reinforce that performance-aware dynamic selection helps mitigate the challenges posed by heterogeneous data distributions, leading to more robust global model convergence.

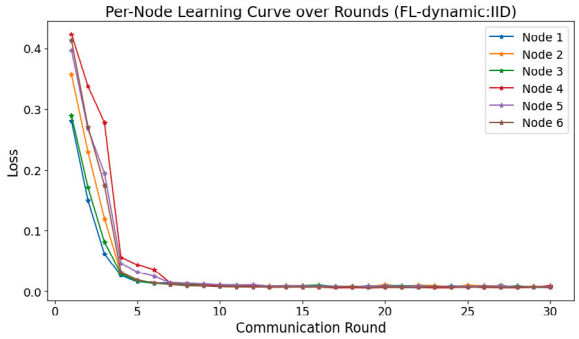
Table 4 summarizes the model performance under non-IID settings, where data distributions across nodes vary. CL is excluded in this setting, as centralized aggregation is not applicable when data remains distributed across clients. Even in this more challenging scenario, FL-dynamic maintains superior performance across all metrics. Specifically, it lowers MAE and RMSE while improving R^2 compared to FL-full. Despite the added complexity of client selection, the training time for FL-dynamic remains nearly identical to FL-full (224.9 vs. 221.3 s), further validating its robustness and practicality.

5. Conclusions

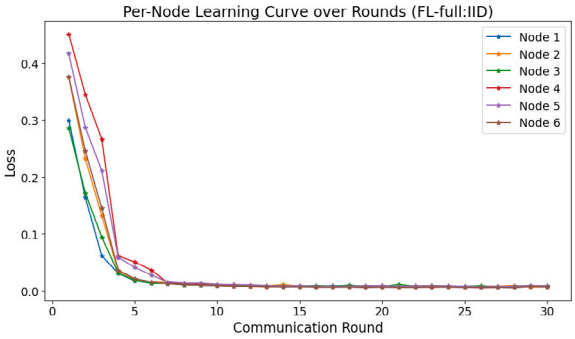
Safety issues have long plagued the construction industry, particularly in dynamic and high-risk outdoor environments. This paper proposed a novel approach for multi-worker safety prediction by integrating Federated Learning (FL) and Digital Twin (DT) technologies. The study utilized real-world construction safety data collected from edge devices and IoT sensors at outdoor sites. To practically validate the framework, we implemented a physical testbed using Raspberry Pi devices as FL clients representing workers and IoT nodes, coordinated by a DT platform and central FL server running on a high-performance PC.

DT served as the core virtualization platform, providing a comprehensive framework for worker safety monitoring. The system supported device registration, model creation, real-time control, visualization of FL training, and safety prediction for individual workers. A DT-assisted application was deployed on physical FL clients, enabling virtual synchronization and control of the edge devices. Initially, the DT platform defined the architecture of the global model, which was then distributed to FL clients for collaborative training. Each FL client fine-tuned the model locally, with updates aggregated centrally to form an optimized global model. To enhance model convergence, a dynamic node selection mechanism based on normal distribution was implemented at each aggregation round, ensuring high-performing clients contributed more effectively to the global model. The resulting model was deployed across clients to enable real-time safety predictions using live streaming data from environmental and worker-based sensors.

In summary, under IID conditions, the proposed FL-dynamic node selection method achieved an MAE of 0.0478, RMSE of 0.0582, and an R^2 score of 97.8%, closely aligning with Centralized Learning (CL) while outperforming the FL-full approach. Under non-IID conditions, FL-dynamic maintained strong performance with an MAE of 0.0578, RMSE of 0.082, and an R^2 score of 96.5%, clearly surpassing FL-full in heterogeneous data environments. In both settings, FL-dynamic also demonstrated favorable training latency, only marginally higher than

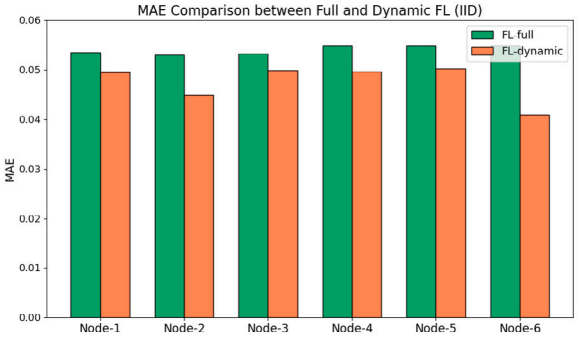


(a) Per-node learning curves for FL-dynamic (IID).

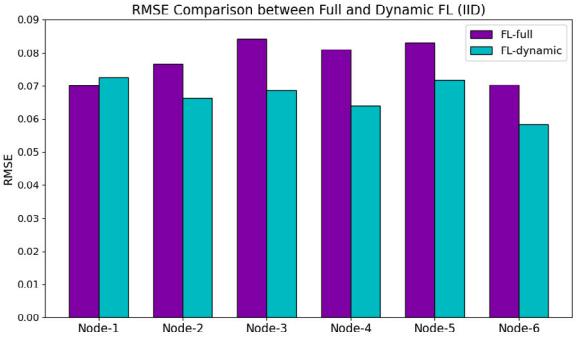


(b) Per-node learning curves for FL-full (IID).

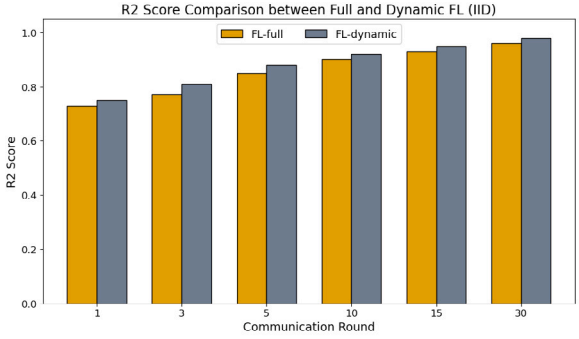
Fig. 25. Per-node validation loss over communication rounds for (a) FL-dynamic and (b) FL-full strategies under IID data conditions.



(a) MAE comparison between full and dynamic FL.

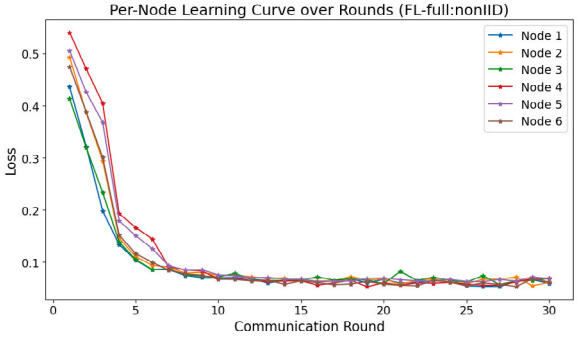


(b) RMSE comparison between full and dynamic FL.

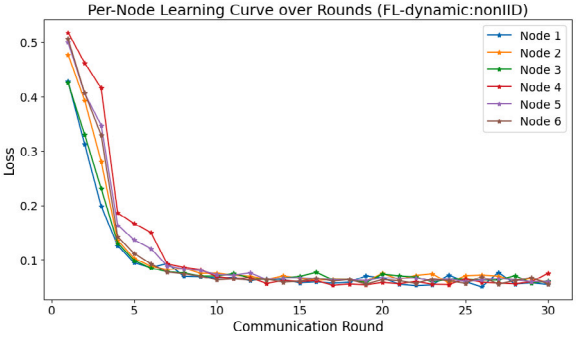


(c) R^2 comparison between full and dynamic FL.

Fig. 26. Evaluation of (a) MAE, (b) RMSE, and (c) R^2 score for full vs. dynamic node selection FL across communication rounds.

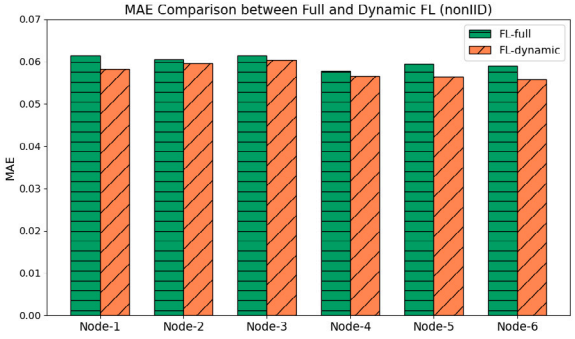


(a) FL-full under non-IID setting.

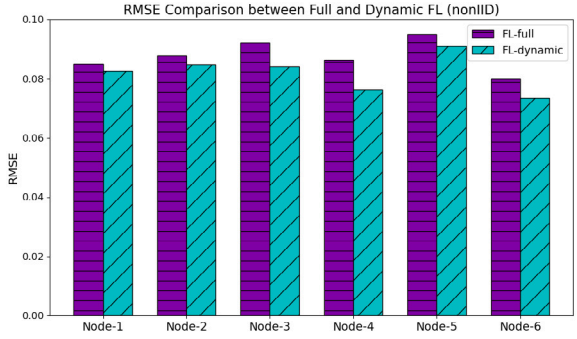


(b) FL-dynamic under non-IID setting.

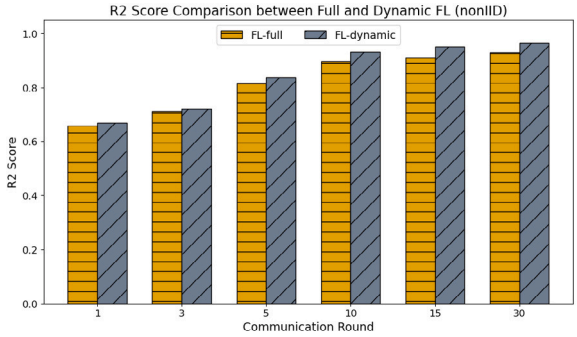
Fig. 27. Per-node learning curves for FL with (a) full participation and (b) dynamic node selection under non-IID data conditions.



(a) MAE comparison under non-IID setting.



(b) RMSE comparison under non-IID setting.



(c) R^2 score comparison under non-IID setting.

Fig. 28. Comparison of FL with full participation and dynamic node selection under non-IID data conditions: (a) MAE, (b) RMSE, and (c) R^2 score across communication rounds and nodes.

FL-full due to node selection overhead, but still well within practical limits. Additionally, the system provided real-time inference with an average delay of 0.551 s, confirming its viability for deployment in dynamic outdoor construction environments.

To the best of our knowledge, this is the first study to develop and experimentally validate a proactive, real-time worker safety monitoring system combining DT and FL in a practical testbed with physical edge devices, targeting multi-worker outdoor construction scenarios. While this study focused on general worker safety prediction, the framework can be extended to model specific accident scenarios (e.g., crane collapses, fires) by integrating additional sensors and specialized training data. Developing such targeted models is an important direction for future work.

For future work, we plan to expand our framework by collecting and integrating additional real-world data features, experimenting with various construction environments, and conducting comparative studies with centralized and alternative FL aggregation schemes to further optimize performance. We also aim to explore additional evaluation metrics such as fairness across clients, long-term robustness under variable site conditions, and responsiveness to rare or catastrophic events through event-specific predictive modeling.

CRediT authorship contribution statement

Sa Jim Soe Moe: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Atif Rizwan:** Validation, Resources, Conceptualization. **Anam Nawaz Khan:** Writing – review & editing, Validation, Investigation, Formal analysis. **Rongxu Xu:** Resources, Investigation, Data curation, Conceptualization. **Do Hyeun Kim:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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